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ZOO504 - Wildlife

Current paper solved question mid term December 2019

1. Wildlife benefits for ecological balance.2 marks

Scavengers (Vultures, eagles, jackals, hyaenas etc.) & decomposers serves to perform important tasks.

1. Nutrients recycling
2. Humus formation
3. Cleaning of environment
4. Soil erosion can be prevented by wild plant cover.
5. Mixing of litter by movement of wild animals.

2. National parks.3 marks

National Park is an area of outstanding scenic merit and natural beauty where the landscape, flora and fauna are protected and preserved in a natural state.

• Public access for recreation, education and research is provided.

• Access roads and other facilities should be planned so they do not conflict with main objectives of National Parks.

3. National Parks Name any three? Name of Four National Parks? 3 marks

01. Margallah Hills National Park.
02. Hazarganji Chiltan National Park:
03. Hingol National Park
04. Kirthar National Park:
05. Ayubia National Park:
06. Chitral Gol National Park:.
07. Sheikh Badin National Park
08. Saiful Maluk National Park.
09. Lulosar Dodipat National Park:
10. broghil valley national park.

3. Seed dispersal.3 marks

Dispersal of seeds------ (squirrels, birds, rodents etc.) Seeds can stick to fur . Squirrels or birds pick them up and drop or bury them. Seed eaters also disperse the seeds by defecation . Caching behavior of many rodents and some birds.

4. Location of Indus dolphin present.2 marks

The Indus Dolphin Reserve is a natural wetland and situated in the Sukkar. From the Sukkur Barrage upstream to the Guddu Barrage near Kashmor, Northern Sindh.

5. Aesthetic and recreational value of wildlife (5 MARKS

It is the pleasure to enjoy the beauty of wildlife

- People enjoy seeing plants and animals in their natural habitats for recreation.
- They are a source of joy, wonder, and pleasure for many people.
- Wildlife tourism (ecotourism).
- Bird watching, is now a growing part of wildlife tourism.
- Animals photographed for recreational purposes also have aesthetic value.
- Exotic game ranches
- Viewing non-native species in a “near-native” habitat.
- Visiting Safari parks/zoo as recreational activity
- Visiting hilly areas for recreational purpose

6. Role of bird and animals in maintain ecosystem

Wildlife play a very integral role in maintaining the fragile balance that ensures the wellness of the earth.

- Each species functions with a specific role – predator, prey, decomposer etc. and hence ecological balance is maintained.
- Certain wild animals and bird are natural Scavengers such as kites, Vultures, eagles, jackals, hyaenas etc.)

• These help to maintain hygienic conditions by

1. Nutrients recycling
2. Humus formation
3. Cleaning of environment

7. Primary objective of Astore Sanctuary. (2)marks

Primary objective

The sanctuary is home to a small population of the near threatened species, Markhor. Markhor is National animal of Pakistan. It is a large species of wild goats unique to this region of the world

7. Astore Sanctuary

Astore sanctuary was announced on 22 Nov 1975. This sanctuary is contiguous to Baltistan Wildlife Sanctuary

Location:

- Astore District
- Between Nanga Parbat
- Gilgit–Baltistan, Pakistan

8. Step take by Ministry of interior for historic areas long.

The Ministry of Interior is the responsible authority for National Parks. The Ministry may establish a National Planning Commission to designate, alter or abolish areas for National Parks. The Commission may review National Park management plans.

National Parks are meant for

- Preserving the nation's unique natural scenery
- Protection of wild fauna and flora
- Preserving historic sites
- Providing public recreation
- Areas for scientific research.
- Areas having unique landscapes, significant ecological systems, or habitats
- Biodiversity that are representative of the natural heritage of the nation.
- Administrative headquarters shall be established at each National Park.

9. What are restrictions on import and export of animals. 3

Restrictions on Import and Export of Animals, Trophies or Meat

No person shall

- Receive by gift, purchase or otherwise any animal of a kind specified in the Second Schedule
- Unless a valid certificate of lawful possession in respect thereof.
- Restrictions on Dealing in Animals, Trophies and Meat
- Wildlife Sanctuary
- National Park
- Game Reserve

10. Causes of depletion of wildlife.5 MARKS

□□□□□**The main factors of Persecution are:**

Wildlife Trade

Trading of wildlife and its products. e.g. Bush meat, Ivory, hide, trophy, fur etc.

Trapping of Wild Animals

- • Pets(cats etc.)
- • Sold for Medical experimental studies (Monkeys, Rabbits etc.)
- • For street shows, circuses (Bears,Parrots,Lions)

Poaching

- • Illegal killing for tusks, oil, decoration (as stuffed preparation), Plumage.
- • Illegal hunting for bush meat, trophies.

Absurd killing

• Tigers, jackals, wolves etc. are killed by the farmers.

□□□Habitat loss poses the greatest threat to species. Habitat destruction is identified as a main threat to 85% of all species described in the IUCN's Red List. The world's forests, swamps, plains, lakes, and other habitats continue to disappear as they are harvested for human consumption.

Forest Loss and Degradation

Expansion of agricultural land, intensive harvesting of timber, wood for fuel etc. is the cause of forest loss. Palm oil plantations in the tropical regions of Africa, Latin America, and Asia have led the large scale destruction of important habitat for many species (Orangutans, tigers, elephants, rhinos). In Malaysia and Indonesia where large tracts of rainforest are cleared to grow palm oil crops. It is an estimate that almost 70% of deforested areas are converted to agricultural land.

Coastal and Marine Areas

Population growth, urbanization, and industrialization results in marine and coastal degradation. Poverty, consumption and land-use patterns contribute to the degradation of marine habitats and to the destruction of the species that rely on them to survive.

Drainage of Wetlands
Wetlands are being drained for cultivation and building of roads etc.

Pollution of various kinds have adversely affected wild animals.
• Insecticides

There has been a considerable damage to birds by insecticides spraying.
• Oil Spillage

Oceanic birds are killed by oil spillage from oil tankers in the sea.
• Noise Pollution

It causes disturbance of hormonal and reproductive behavior of many wild animals.
• Water Pollution

Domestic as well as industrial waste dumped into water bodies causes decline in stocks of fishes.
• Animals like mussels have been unfit for human consumption

• Various natural calamities like floods, hurricane, fire etc. have also caused massive destruction of wildlife.

11. causes of degradation of wetlands.

The primary pollutants causing wetland degradation are
• Sediments
• Fertilizers
• Human sewage
• Animal waste
• Road salts
• Pesticides
• Heavy metals

12. Location of hingol national park 2m

It is located along the Makran coast in southwestern, Balochistan-Pakistan. It lies within three districts of Balochistan, Lasbela, Gwadar and Awaran.

13. kinds of flora in ayubia national park..

Major Flora:
The park holds 104 species of plants. The main floral species are
• Cedrus deodara
• Blue pine
• Yew
• Silver fir
• Horse chestnut
• Oak

14. Give Primary objective of Protected Area with Sustainable use of Natural Resources (5)

Protected Area with Sustainable Use of Natural Resources
Definition:
Protected areas that conserve ecosystems and habitats, together with associated cultural values and traditional natural resource management systems.
Generally large, with most of the area in a natural condition. Where a proportion is under sustainable natural resource management. Where low-level non-industrial use of natural resources compatible with nature conservation is seen as one of the main aims of the area.
Primary objective
• To protect natural ecosystems
• Use natural resources sustainably, when conservation and sustainable use can be mutually beneficial.

Other objectives
• To promote sustainable use of natural resources.
• To promote social and economic benefits to local communities.
• To facilitate scientific research and environmental monitoring.

To facilitate recreation and appropriate small-scale tourism.

15. I.W.C is mandated, to cooperate with other intergovernmental organizations including:5

□□□□**I.W.C is mandated, to cooperate with other intergovernmental organizations including:**

• International Maritime Organization (IMO)
• Regional Marine Fisheries Organizations (RFMOs)
• Marine Council
• UNEP

Collaborations:

Collaboration with other international organizations on some issues include:

• Climate change
• Marine debris
• Marine safety
• Habitat degradation
• Marine Protected Areas
• Marine Science

Headquarters:
The headquarters of the I.W.C is in Impington, near Cambridge, England. The Commission has three main committees:

• Scientific
• Technical
• Finance and Administration

Purpose:
The purpose of the I.W.C as specified in its constitution is
"Safeguarding for future generations the great natural resources represented by the whale stocks;"
Original members consisted only of the 15 whale-hunting nations.

16. Note on Hub dam wildlife sanctuary. 3

□□Hab Dam Wildlife Sanctuary was declared in July 1974. It is Pakistan's third largest dam

Area:
27,219ha

Location:
It is situated in the north of Karachi bordering the Kirthar National Park Balochistan

Features:

• Hab Dam Wildlife Sanctuary is covering an area of 32 square miles
• Much of the shoreline is steeply shelving and stony
• Many shallow bays
• Small islands
• The greater part of the reservoir (in Balochistan) is unprotected
• The Eastern shore and of the Dam (in Sindh) is protected in the Kirthar National

Fauna:
An ideal place for bird watching
Waterfowl (both resident and migratory). The lake also provide refuge to migratory birds

▪ Cranes
▪ Pelicans
▪ Ducks
▪ Waders

Fauna:
The surrounding hills are home for

• Urial
• Sindh Wild Goat
• Chinkara Gazelle
• Pangolin
• Wolf
• Jackal
• Common fox

• Numerous reptiles

17. name verities of living organisms in wildlife. 3 marks

Organisms that grow or live wild in an area without being intervened by humans.

Wild life is natural renewable resource with a great importance throughout world.

It comprises innumerous varieties of living organism in their natural habitats.

1. Wild Plants
2. Wild Animals
3. Fungi
4. Microorganisms

18. Wetland Definition?2

A wetland is a place where the land is covered by water, either salt, fresh or somewhere in between. Marshes and ponds, the edge of a lake or ocean, the delta at the mouth of a river, low-lying areas that frequently flood—all of these are wetlands.

19.What is gene pool.2 marks

The gene pool is the set of all genes, or genetic information, in any population, usually of a particular species

Gene pool refers to the total number of **genes** of every individual in a population. It usually involves a particular species within a population. ... A large **gene pool** indicates high **genetic** diversity, increased chances of **biological** fitness, and survival.

20. How many Ramsar sites are present in Punjab write name.2 marks

In 2013, 19 (nineteen) Ramsar sites has been declared in Pakistan.

Punjab

1. Uchhali Complex
2. Taunsa Barrage
3. Chashma Barrage

21. What is the IRDCP project.5 marks

Indus River Dolphin Conservation Project (IRDCP)

introduction:

Indus River Dolphin *(Platanista gangetica minor)*, an endangered species endemic to the Indus River system in Pakistan. This project was developed to conserve a viable population

Habitat: Habitat has been reduced to one fifth of its historical range and further degraded due to shortage of water.

Population: Divided in sub-populations because of the six barrages constructed on the River Indus.

Project Area: Covering an area of 200 km between Guddu and Sukkur barrages in the province of Sindh and some extensions of the project covers the Punjab.

Project Objectives:

• Protecting the innate biodiversity of the Lower Indus River Basin Eco-system.

• Ensuring the sustainable use of riverine biodiversity.

• Promoting actions to mitigate pollution and the wasteful extraction of riverine resources.

Project Sponsors:

• WWF-Switzerland

• Swedish International Development Agency (SIDA)

• Engro-foods Limited, Pakistan.

Stakeholders:

• WWF-Pakistan

• Sindh Wildlife Department (SWD)

• Agriculture Extension Department, Sindh (AED)

• Sindh Environmental Protection Agency (SEPA)
• Sindh Fisheries Department (SFD)

Project Approach:
The Indus River Dolphin Conservation Project focuses on the root causes of biodiversity loss by linking the protection of the Indus River Dolphin with measures in the agricultural and fisheries sectors.

22. What is main objective of the national park.5 marks

Definition:
Large natural or near natural areas set aside to protect large-scale ecological processes, which also provide a foundation for environmentally scientific, educational, recreational and visitor opportunities.
Primary objective
To protect natural biodiversity along with its underlying ecological structure.
Other objectives
• To manage the area in order to perpetuate, in as natural a state as possible.
• To maintain viable and ecologically functional populations.
• To contribute to local economies through tourism.
• To manage visitor use for inspirational, educational recreational purposes.

23. Importance of margla hill national park.3 marks

Location:
North of the Islamabad City, Capital of Pakistan.
Area
• 12,000 ha

Features:
• Elevation range 550m to 1500m.
• Rugged topography
• Numerous valleys
• Precipitous slopes
• Margalla hills have beautiful torrents gushing down in the monsoon.
• Natural springs are also present.
• The Park includes the Margalla Hills which form's the foothills of the Himalayas, along with Shakarparian Park and Rawal Lake.

Snag & cavity trees?

Snags:

- Dead or partially dead standing trees. Nesting sites for forest-dwelling bats/birds.

Cavity Trees:

- Holes excavated in snags by woodpeckers. Shelter and nesting cover by many species.

24. Define protected area.3 marks

Protected Area

“ A clearly defined geographical space, recognised, dedicated and managed, through legal or other effective means, to achieve the long-term conservation of nature with associated ecosystem services and cultural values”.

Name game reserve of the federal territory. 2

Federal territory: have one game reserve 1

Federal Territory

Islamabad

25. Major Challenge of vaterniarans.5 marks

• Environmental contaminants
• Infectious diseases

• Outbreaks that potentially threaten wild and domesticated animals as well as humans.
Wildlife veterinarians are necessary and should made available to care of wild animals.

Veterinarians are to prevent
• Infectious disease transfer at the domestic animal-wildlife interface.
• Movement of diseases between wildlife and domestic species.
• Reducing contact between domestic and wild animals where necessary.

26. Home range

Home Range: where an animal lives and moves on a periodic basis

27.Game count

➢ Game count to be conducted as it provides an estimation of the game population
➢ It is to ascertain the number of female animals, as this indicates the production potential.
➢ Knowing game count statistic informs the farm's threshold and when, what and how much must be hunted.
➢ Managing erosion and conserving pasture are also crucial aspects.
➢ It is an estimate that walking 6km to water isn't unnatural for an animal.
➢ Some of the Reserve's water points will be done away or moved to more suitable locations.

Name 2 wildlife sanctuaries of KPK. 3

Khyber PakhtunKhwa(KPK)

- Manglot
- Argam Basti

28.Which activities prohibited in national park

The following activities shall be prohibited within the National Parks:
• Burning of vegetation or setting fires to clear land.
• Hunting animals or catching fish.
• Polluting water or air
• Picking or removing flower or any other vegetation.
• Engraving, sketching or defacing trees, bark, stone or signs.

What is monitoring trade in CITES. 5

Convention on International Trade in Endangered Species (C.I.T.E.S)

Monitoring trade

- The backbone of CITES is the permit system that facilitates international cooperation in conservation and trade monitoring.
- Permits are issued only if a country's Management and Scientific Authorities determine that trade is legal and does not threaten the species' survival in the wild.
- These permit allows inspection at ports of export and import to quickly verify that CITES specimens are properly documented.
- This trade monitoring has created an information on the management and use of CITES species worldwide.
- CITES has helped global conservation of species.
- CITES provides tools to effectively conserve the world's diverse natural resources.
- It control global over-exploitation of wildlife.

29.Exotic species.2 marks

An introduced species, alien species, exotic species, foreign species, non-indigenous species, or non-native species is a species living outside its native distributional range, but which has arrived there by human activity, either deliberate or accidental.

Game reserve .2 marks

Game Reserve: An area where regulated hunting & shooting is allowed

Define game reserve. 3 marks

Definition:

"A game reserve is an area wherein controlled hunting and shooting is permitted on permit basis*"*

A game reserve (wildlife preserve) is a large area of land where wild animals live safely or are hunted in a controlled way for sport.

1. In the game reserves the major focus is specifically the animals.
2. If hunting is prohibited, a game reserve may be considered a "Nature Reserve"
3. Wherein all aspects of naturally-occurring life in the area are considered.
4. Most of the areas in game reserves have created to provide habitat protection for animal species commonly referred to as game (huntable species for sport or meat)

30. Short note on ayobia national park.3 marks

□□□Ayubia National Park was declared a National Park in 1984. Ayubia was named after Muhammad Ayub Khan, the then President of Pakistan.

Location:

Abbottabad District, Khyber Pakhtunkhwa- Pakistan.

Area:

Initial area was 1684 ha, expanded through northern extension in 1998. Total area 3,312 hectares (8,184 acres)

Features:

• This Natiopnal park is surrounded by three small towns(Nathia gali, Ayubia and Khanaspur)
• Rainfall: 1,244 cm
• Temperature: 3 °C - 11 °C

The Area Supports

• Temperate coniferous forest
• Temperate broadleaf and mixed forest
□□□
• These forests have an average elevation of 8,000 feet (2,400 m) above the sea level.

Major Flora:

The park holds 104 species of plants. The main floral species are

• Cedrus deodara
• Blue pine
• Yew
• Silver fir
• Horse chestnut
• Oak

Plants belonging to 19 families are known for their medicinal properties.

Major Birds Fauna:

• The park harbors up to 203 species of birds.
• Golden eagle
• Vulture
• Eurasian sparrow
• Hawk
• Hill pigeon
• Khalij Pheasant (30 species)
• Spotted dove
• Collared dove

Major Mammal Fauna:

Ayubia National Park supports 31 species of mammals.

• Asiatic leopard
• Hill fox
• flying squirrel
• Masked civet
• Rhesus Macaque

31. Note on chitral park.

Chitral Gol National Park is established in 1984. “Gol” is local langauge meaning “Valley”.

Location:

Hindukush range. Beside the Chitral River in Chitral District - Khyber Pakhtunkhwa-Pakistan.

Area:

7,780 ha **Features:**

- Rainfall is estimated to be 462 ml.
- Weather is cold and dry.
- The temperature ranges from -12.2 to 43.3 °C.

Major Flora:

- The park is rich in a particular tree. Cedar **Major Fauna:**

- Chitral National Park provide shelter to a vast biodiversity, especially Markhor.
- Holds the largest population of in the world.
- Siberian ibex
- Ladakh Urial
- Asian black bear
- Snow leopard (Non-permanent resident)
- Tibetan wolf
- Red fox
- Yellow-throated marten
- Himalayan otter

Birds Fauna:

- Bearded vulture
- Himalayan vulture
- Demoiselle crane
- Peregrine falcon
- Golden Eagle
- Himalayan snow cock
- Himalayan monal
- Snow partridge
- Rock partridge

32.Constitution members.3marks

Constitution of Board

The Board shall consist of

Chairman

Vice-Chairman

Number of members as may be appointed by Government.

Chief Minister of the Punjab shall be the chief warden and *ex-officio* Chairman of the Board.

Minister incharge of the Department shall be the *ex-officio* Vice Chairman of the Board.

Secretary to Government, Forestry, Wildlife, Fisheries and Tourism Department shall be the *ex-officio* Secretary of the Board.

2- Mention places declared bt IUCN categories wise? 5

1. Category I
   - O Category Ia
   - O (Strict Nature Reserve)
   - O Category Ib (Wilderness Area)
2. Category II (National Park)
3. Category III (Natural Monument or Feature)
4. Category IV (Habitat/Species Management Area)
5. Category V (Protected Landscape/\Seascape)
6. Category VI (Protected area with sustainable use of natural resources)

33. Sanctuary

Definition:

Wildlife Sanctuary, is a protected area, provides an undisturbed breeding ground for the protection of wildlife. It is a naturally occurring sanctuary, may be an island.

It provides protection for species from

• Hunting
• Predation
• Competition

These wildlife refuges are generally officially designated territories. Such refuges can protect animals that are endangered.

34. Kirthar park

Location

The **Kirthar National Park** is situated in the Kirthar Mountains in Karachi and Jamshoro District in Sindh, Pakistan. South –Western part of Sindh.

Establishment

- • It was founded in 1970 and stretches over 3,087 square kilometres.
- • Second largest National Park in Pakistan after Hingol National Park.
- • Hub lake with 82 square km

Major Fauna

26 principal species of mammals are believed to occur in the park.

- • Indian leopards
- • Striped hyenas
- • Indian wolves
- • Urials
- • Chinkara gazelles
- • Sind wild goats(Rarely)

Birds fauna is equally rich with at least 58 varieties.

- • Kingfisher
- • Larks
- • Eagles
- • Vultures

This park is primarily established to protect and preserve wild goats and sheep of Sindh.

- • 1971-1200
- • 1977-2500
- • Todate-4000

Blackbuck antelopes are kept in enclosures for a reintroduction project at Kirthar National Park. The park is accessible by four wheel drive vehicles Two rest houses belonging to the Sindh Wildlife Department are available for tourist accommodation.

35. Zoological garden

Based on area, Zoological garden can be categorized into four Zoos

• Large Zoo
• Medium Zoo
• Small Zoo
• Mini Zoo

36. Ssc abbreviation

Species Survival Commission (ssc

37. Hub dam area

Area:

27,219ha

3- Negative impact of predation?

Negative impact of wildlife occurs when growing human populations overlap with established wildlife territory.People lose their crops, livestock, property, and sometimes their lives.

38. Name the largest national park of Pakistan 2 marks

hingol national park is largest national park of Pakistan.

4- Who is Carl Hagenbeck? 2

German collector, Carl Hagenbeck, built the first wild animal park in 1848 • He allowed the animals outdoor access, believing their enclosure should more closely resemble nature • The Zoological Society of London justified London ’s zoological collection as “The advancement of zoology and animal physiology and the introduction of new and curios subjects of the Animal Kingdom.

Purpose of IWC (2

International Whaling Commission (I.W.C)

The purpose of the I.W.C as specified in its constitution is
"Safeguarding for future generations the great natural resources represented by the whale stocks;"
Original members consisted only of the 15 whale-hunting nations

39.What is power to seizure in wildlife.3marks

Any officer authorized by Government in this behalf, may seize any wild animal, dead or alive anything whatsoever used or suspected to have been used in the commission of an offence under this Act.
• Duty to produce permit on demand made by any officer or person authorized in this behalf.
• Duty of coupe purchaser, chowkidar, village watchman

40.Acts prohibited under law of national park.5marks

□□□□□**Acts prohibited under Law in National Parks:**
• Firing of guns
• Hunting, killing, trapping or capturing wild animals inside or within a radius of three miles of its boundaries.
• Interfering with animals and plants
• Clearing land for cultivation.
• Flow of polluted water is also prohibited. Under the regulations, these